PROJECT PROPOSAL

FAST - National University of Computer and Emerging Sciences

Karachi Campus | FAST School of Computing | Department of AI and Data Science

AL-2002 Artificial Intelligence Laboratory | Fall 2026 | Section BCY-5A

Instructor: Miss Amma Aslam | Proposal weightage: 1 mark

Group Information

Position	Name	Roll number	Section	Planned role
Leader	Muhammad Hasnain Memon	24K-2001	BCY-5A	Architecture, models and explanations
Member 2	Akbar Hussain	24K-2026	BCY-5A	Data, clustering and interface

Field	Value
Project title	Explainable Fake News Headline Analyzer
Project idea	Idea No. 19 selected from the official list; not an own idea
Category	Part B - Applied and Data Projects
GitHub repository URL	https://github.com/HasnainMemon04/fake-news-headline-analyzer
Date of submission	Pending instructor announcement; prepared 17 October 2026

The team has two members and complies with the official group-size limit. Idea allocation on the BCY-5A group sheet must be confirmed.

1. Project Title and Summary

Explainable Fake News Headline Analyzer will classify English headlines as likely real or likely fake based on a labelled dataset. Students and readers will enter a headline and inspect the prediction and influential words. Two supervised classifiers will learn language patterns that fixed keyword rules cannot reliably describe. Topic clustering will organize similar headlines for exploration. The application will support learning and critical reading, while making clear that a headline classification does not independently establish factual truth.

2. Problem Statement and Motivation

Readers encounter headlines without the evidence needed to verify their claims. Short text can omit context, use irony, or repeat the wording of a genuine report, so suspicious words alone cannot determine whether a claim is true. Manually checking every headline requires opening articles, locating original sources, and comparing accounts, which is difficult when many items arrive together. Students and online readers often rely on familiar publishers, search engines, or fact-checking services, but these approaches require time and judgement. This project addresses a narrower, measurable task: predicting the real or fake label assigned to a headline in a public dataset. Vocabulary combinations and uncertain labels make fixed rules inadequate, while sparse text features allow manageable statistical learning. Explanations will reveal the words affecting predictions, and topic exploration will help expose patterns and bias in the data. Building this system is worthwhile because it connects supervised learning, clustering, and honest evaluation to a familiar information problem.

3. Problem Formulation

3.1 PEAS Description

Component	Description for this project
Performance measure	Primary macro-F1; accuracy, per-class precision, recall and F1, confusion matrices, fit and prediction time.
Environment	Labelled English headlines for offline learning; unseen headlines entered into a local interface for inference.
Actuators	Display predicted dataset class, model score, signed influential terms and topic assignment.
Sensors	Headline text; reference labels available only during training and evaluation, never as inference inputs.

3.2 Environment and Agent Type

Property	Classification and interpretation
Observability	Partially observable: headline visible, supporting evidence and factual truth hidden.
Determinism	Stochastic task: text does not uniquely determine truth; a fitted inference pipeline is deterministic.
Change over time	Static during a prediction; language and news distributions can drift between datasets.
State and actions	Discrete text inputs and label actions; internal TF-IDF weights are continuous.
Agents	Single-agent; no strategic opponent.
Agent type	Learning agent: offline fit updates parameters using labels; frozen models predict at runtime.

3.3 Problem Representation

Element	Representation
State representation	Validated headline transformed to a sparse, non-negative TF-IDF vector x.
Actions / moves	Select y in {0: real, 1: fake}; return explanation and topic.
Initial state	Unclassified headline with a saved vocabulary and trained model.
Goal or terminal test	Produce a label; during evaluation compare it with the held-out dataset label.
Cost, utility or evaluation	Logistic log loss or Naive Bayes likelihood during fit; macro-F1 on unseen records for comparison.

4. Proposed AI Approach

#	Technique	Role	Own code or library
1	TF-IDF + Multinomial Naive Bayes	Probabilistic classifier	scikit-learn; team pipeline
2	TF-IDF + Logistic Regression	Linear classifier and explanations	scikit-learn; team explanation code
3	K-Means	Unlabelled topic exploration	scikit-learn; team interpretation
B	Majority-class baseline	Minimum comparison	DummyClassifier, most_frequent

TF-IDF represents short headlines with sparse word weights, keeping training practical on a laptop. Multinomial Naive Bayes combines class priors with word likelihoods under a conditional independence assumption. Logistic Regression learns a weighted sum of features and maps it to a class probability; this provides a contrasting decision rule. For class fake, each term contributes x_j times w_j to its log odds, with the intercept reported separately. Naive Bayes explanations use x_j times the difference between the two class log likelihoods, plus the log prior ratio. Highlighted terms therefore explain the selected model, rather than arbitrary keywords. K-Means minimizes within-cluster squared distance to explore lexical topics without using truth labels. A majority-class predictor tests whether learned patterns improve on simply choosing the common training label.

4.1 Planned Class Design

NewsDataset: private records/split IDs; load(), clean(), split(). **HeadlineAgent:** abstract base; protected vectorizer/model; fit(), predict(), explain(). **NaiveBayesAgent** and **LogisticAgent** inherit it and override fit() and explain(). **TopicClusterer:** private centroids; fit(), assign(), top_terms(). **ExperimentRunner:** manifest/seed; compare(), export(). Controlled methods encapsulate state; the UI uses the shared agent API.

5. Scope

#	In-scope feature	Why it is essential
1	EDA of lengths, common words and available source metadata	Meets Idea 19; documents missing data, outliers and bias.
2	Naive Bayes and Logistic Regression on identical test records	Measures two supervised approaches against the baseline.
3	Highlight signed terms behind each model prediction	Makes decisions inspectable; handles no known terms explicitly.
4	K-Means topics with top terms and sample headlines	Demonstrates and interprets unsupervised learning.
5	Local Streamlit headline box and analysis views	Connects predictions, explanations, EDA and topic exploration.

Out of scope: Article-body classification, automated fact verification, live scraping, multilingual models, deep learning, paid APIs and deployment. Model probabilities will be labelled uncalibrated scores, not certainty of truth.

6. Data or Environment Source

Field	Verified source and planned use
Source name and URL	ISOT Fake News Dataset, University of Victoria [1]; verified Kaggle mirror [3]; university archive [2].
Type	Public dataset; locally loaded CSV files.
Access requirements	Mirror downloaded anonymously; no key or account used. University ZIP is an alternative; local caching avoids repeated downloads.
Approximate size	Verified 44,898 articles: 21,417 real and 23,481 fake; 38,720 unique normalized titles after duplicate removal.
Required columns	Verified title, text, subject, date. Use title only; derive labels from True.csv/Fake.csv. No publisher column is supplied.
Licence / terms	Mirror metadata lists CC BY-NC-SA 4.0. Cite original authors; the university description states no separate explicit licence. Keep article text local.
Offline fallback	Verified CSVs cached in data/raw with SHA-256 records; split manifest in data/processed. Future trained artifacts stay local in outputs. Prepared runs require no network.

Preparation: Normalize Unicode/case/punctuation for duplicate checks; remove empty and conflicting titles and exact normalized repeats. Group near duplicates by token-set Jaccard similarity of at least 0.90 before splitting. Inspect length outliers and document rules without tuning to test outcomes. Source EDA will audit documented Reuters collection provenance and unknown fake publishers, with missing-source coverage explicit. Subject and CSV filename are not publishers; this dataset cannot support a publisher-disjoint evaluation. Full local schema/count/hash checks and a separate feasibility spike are recorded in repository documentation.

7. Evaluation Plan

Requirement	Planned experiment
What will be compared	Multinomial Naive Bayes, Logistic Regression and majority baseline with the same representation and test IDs.
Metric(s)	Accuracy; per-class precision, recall and F1; macro-F1; confusion matrices; fit and batch prediction seconds on the same hardware.
Baseline	Most frequent training class; tie rule and label order recorded.
Test instances / test set	Five-fold stratified group splitter: reserve its first fold (7,744 titles); 30,976 development titles. No duplicate group crosses splits; final test remains unevaluated.
Number of runs and seed	random.seed(42), NumPy seed 42, estimator/split random_state=42. Three-fold grouped stratified CV within training only; one final test evaluation per selected model.
Planned charts	Labelled model metric bars; confusion matrices with class names; length distributions; frequent terms and source coverage; cluster sizes and top terms. Include a numeric results table.

Fit TF-IDF inside each CV fold: word unigrams, 20,000-feature cap, min_df=2 and English stop words. Compare NB alpha in {0.5,1} and LR C in {0.5,1,2}; refit on development data only. Fit K-Means without labels; choose k from {3,5,8} using sampled silhouette and readable terms, with seed 42 and n_init=10. Report cluster sizes, top terms and representative headlines. Feasibility checks do not replace final test results; no accuracy threshold is promised.

8. Division of Work

Member	Modules / components owned	Approximate share
Muhammad Hasnain Memon - 24K-2001	Architecture, preprocessing integration, classifiers, word explanations, evaluation and repository coordination	50%
Akbar Hussain - 24K-2026	Dataset acquisition, EDA, K-Means, visualizations, Streamlit interface and user documentation	50%

Both members will collaborate on integration, experiments, testing and final viva preparation. Each will write and explain their modules, make genuine commits from their own account and review one integration boundary. The final report will record actual contributions and cite external code, datasets and AI assistance.

9. Milestones and Timeline

#	Demonstrable milestone	Target	Owner
1	Validated dataset, source audit, split manifest and EDA charts	Week 9	Akbar; Hasnain integration
2	Two classifiers and baseline reproducibly trained; CV table exported	Week 10	Hasnain; Akbar review
3	Interpreted topic clusters and verified signed word contributions	Weeks 11-12	Akbar + Hasnain
4	Streamlit input, explanations and analysis views run offline; error cases checked	Weeks 13-14	Akbar; both integrate/test
5	Final held-out experiments, three or more charts, report PDF/MD, fresh-clone verification and clean repository	Week 15	Both; leader submits

Week targets are provisional planning slots; the current teaching week and proposal deadline need instructor confirmation. If fewer weeks remain, compress the first four milestones while retaining the Week 15 submission. Individual viva takes place in submission week; evaluation finishes before the Week 16 Lab Final Exam.

10. Expected Outcome and Risks

A working submission will open a local dashboard where users classify English headlines, inspect signed word contributions, and explore topic groups. The report will present measured classifier and baseline scores, confusion matrices, runtime, and interpreted clusters without promising an accuracy threshold. The first risk is learning publisher style or duplicated wording instead of useful headline patterns; duplicate grouping, source auditing, and explicit generalization limits will address it. The second risk is losing access to the data; the verified local cache, checksums and documented mirror will support continuity, with article text kept outside Git. Empty input and missing artifacts will produce actionable messages.

References and access record

[1] University of Victoria ISOT dataset page: https://onlineacademiccommunity.uvic.ca/isot/datasets/

[2] University archive: https://onlineacademiccommunity.uvic.ca/isot/wp-content/uploads/sites/7295/2023/03/News-_dataset.zip

[3] Kaggle mirror: https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset

[4] ISOT official description: https://onlineacademiccommunity.uvic.ca/isot/wp-content/uploads/sites/7295/2023/02/ISOT_Fake_News_Dataset_ReadMe.pdf

Sources and complete mirror CSVs checked 17 October 2026. Cite the dataset papers requested in [4] in the final report. University source documents: AL-2002 Project Guidelines, Project Ideas and Project Proposal Template, Fall 2026, supplied as DOCX.

For office use - Lab Instructor

Field	Reserved for instructor
Proposal status	
Comments	
Signature and date	